

# Ria Banerjee

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## TECHNICAL SKILLS

- *Languages & Frameworks:* Python, React, TypeScript, Django REST, Flask, Git, SQL, Go, .
- *AI/ML & LLM:* LangChain, LangGraph, CrewAI, RAG (Retrieval-Augmented Generation), Vector DBs (pgvector, Supabase), vLLM, MCP servers, Temporal, Claude code, AWS Bedrock.
- *Database & Message streaming:* PostgreSQL, MySQL, Kafka, RabbitMQ, Redis Streams.
- *Cloud Infrastructure:* EKS, ECS, S3, ALB, Kubernetes, Helm, Terraform, Docker, Ansible, Istio, API Gateway.

## WORK EXPERIENCE

### **SWE Backend & Infrastructure, EAGL Technology Inc.**

May 2024 – Present | USA

*Gunshot detection platform deployed across Fortune 500 enterprises, schools, and hospitals.*

- Scaled real-time gunshot detection platform processing 6M events/hour from 100K IoT devices using event-driven microservices (Python, Django REST, Kafka, PostgreSQL) achieving under 250ms p99 latency.
- Built Kafka-to-WebSocket streaming pipeline with Redis pub/sub routing, handling 8K+ concurrent connections across 120+ tenants with 99.9% delivery, replacing batch polling that lagged by 5 seconds.
- Architected RAG agent over operational sensor data using AWS Bedrock + pgvector with MCP-based tool access, enforcing tenant-level RBAC and in-VPC data residency.
- Designed end-to-end observability with Prometheus, Grafana, Fluentbit, OpenSearch, Opentelemetry, Jaeger. Surfaced 25+ bottlenecks and drove 83% latency reduction in critical event processing path.
- Implemented MySQL binlog-based CDC pipeline for air-gapped infrastructure achieving real-time replication, eliminating 4+ hour batch delays and enabling near real-time consistency.
- Deployed AWS infrastructure (EKS, ALB, S3, RDS, Route 53) with Terraform IaC and Helm-based blue-green deployments, enabling zero-downtime across production environments.
- Deployed production RabbitMQ+MQTT cluster on Kubernetes handling 100K+ concurrent IoT connections with automatic failover and message persistence.

### **Founding Software Engineer, BabelCast Inc.**

October 2023 – January 2024 | Remote, USA

*Real-time voice augmentation software.*

- Scaled voice AI platform from 0 to 5K+ concurrent users in 3 months by architecting Django REST microservices with API Gateway, horizontal autoscaling on ECS, and request throttling handling 10K+ API calls daily.
- Designed and implemented authentication system using Keycloak (OIDC) with JWT-based session tokens and RBAC across services.
- Reduced voice model query latency 25% (from 320ms to 240ms) by designing GraphQL API with DataLoader batching, Redis caching layer, and optimized database queries serving 5K+ voice models.

### **System Engineer, Equifax (Tata Consultancy Services)**

May 2021 – August 2022 | India

- Contributed to the development of Spring Boot microservices handling large-scale credit data workflows.
- Worked with Apache Kafka to process streaming data and ensure reliable message delivery between services.
- Supported deployment and monitoring of Java services in Kubernetes-based environments.
- Participated in code reviews and followed enterprise coding and security standards.

## EDUCATION

University of North Carolina Charlotte, USA | Masters in Computer Science

August 2022-May 2024

Nirma University, India | Masters of Computer Applications

August 2019-May 2021

Ahmedabad University, India | Bachelors of Computer Applications

August 2016-May 2019

## RESEARCH

Graduate Research Assistant, University of North Carolina, Charlotte

February 2023-October 2023